

ABOUT

I am an M.S. student in Computer Science at **Institut Polytechnique de Paris (IP Paris)**, with a B.S. in Data Science and Big Data Technology from **Southern University of Science and Technology (SUSTech)**, currently a Research Assistant at **SUSTech-NLP** (advised by Prof. Guanhua Chen).

My research is on **how heterogeneous-capability LLMs should work together**. I started with per-query routing: in **RouterXBench**, I built a three-axis evaluation framework that decouples router intrinsic ability from scenario fit. The same composition question reappears when an agent has to adapt to a new environment over many steps; in **Scope-then-Cover**, I instantiate this as a single explore action where the strong actor scopes a small candidate subspace and frozen cheap workers verify it in parallel, effective in both test-time adaptation (TTA) and online experiential learning (OEL) settings.

EDUCATION

Institut Polytechnique de Paris

M.S. in Computer Science

Paris, France

Sep 2025 – Sep 2027

- **ReflexSQL**: an exploration-and-memory Text-to-SQL agent on Spider 2.0-Lite [[Poster](#)].

Southern University of Science and Technology (SUSTech)

B.S. in Data Science and Big Data Technology

Shenzhen, China

Sep 2021 – Jul 2025

- Bachelor's thesis on LLM-SLM collaborative routing (continued in **RouterXBench**).

PUBLICATIONS

Towards Fair and Comprehensive Evaluation of Routers in Collaborative LLM Systems

- Wanxing Wu, He Zhu, Yixia Li, ..., Guanhua Chen[†]
- Under Review, 2026 [[arXiv:2602.11877](#)]

Can Weak Models Explore? Scope Then Cover for LLM Agent Adaptation

- Wanxing Wu*, He Zhu*, Guanhua Chen[†] (*Equal contribution)
- Under Review, 2026

RESEARCH EXPERIENCE

SUSTech-NLP Group

Research Assistant, advised by Prof. Guanhua Chen.

Shenzhen, China

Jan 2025 – Present

Heterogeneous-capability LLM composition: per-query routing and per-step decomposition in agent loops.

Institute of Intelligent Medical Research (IIMR), BGI

Machine Learning Intern (AI for Genomics).

Shenzhen, China

Jul 2024 – Jan 2025

cfDNA feature representation + ensemble learning for early colorectal cancer detection (**94.88% sensitivity**).

HONORS & AWARDS

- **Outstanding Graduate**, Zhiren College, SUSTech (Jul 2025).
- **Outstanding Graduation Thesis**, Top 5%, SUSTech (Jul 2025).
- Outstanding Student Scholarship, SUSTech (Oct 2024).

SKILLS

LLM & ML: inference pipeline (vLLM, SGLang), agentic-RL workflows, SFT/RL training.

Programming & Dev: Python, PyTorch, Docker, Linux, Git, shell scripting.

Languages: English (IELTS 6.5, CET-4, CET-6); Chinese (Native).